Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at /home/guest/ENVIRON 872/EDE_Fall2025
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
library(ggthemes)
```

```
##
## Attaching package: 'ggthemes'
##
## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```r
library(ggplot2)
here()
```

```
## [1] "/home/guest/ENVIRON 872/EDE_Fall2025"
```

```r
NTL_LTER_processed_Chem_nutr <- read.csv("./Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPa
NEON_NIWO_processed_Litter_mass_trap <- read.csv("./Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Proces

#2
class(NTL_LTER_processed_Chem_nutr$sampledate)
```

```
## [1] "character"
```

```r
class(NEON_NIWO_processed_Litter_mass_trap$collectDate)
```

```
## [1] "character"
```

```r
NTL_LTER_processed_Chem_nutr$sampledate <- ymd(NTL_LTER_processed_Chem_nutr$sampledate)

NEON_NIWO_processed_Litter_mass_trap$collectDate <- ymd(NEON_NIWO_processed_Litter_mass_trap$collectDate
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background

- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
my_theme01 <- theme_classic() +
  theme(text = element_text(family = "serif",
                            size = 8),
        plot.title = element_text(face = "bold",
                                  size = 9),
        axis.title = element_text(face = "bold",
                                  size = 8),
        plot.background = element_rect(fill = "white"),
        legend.position = "right")
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

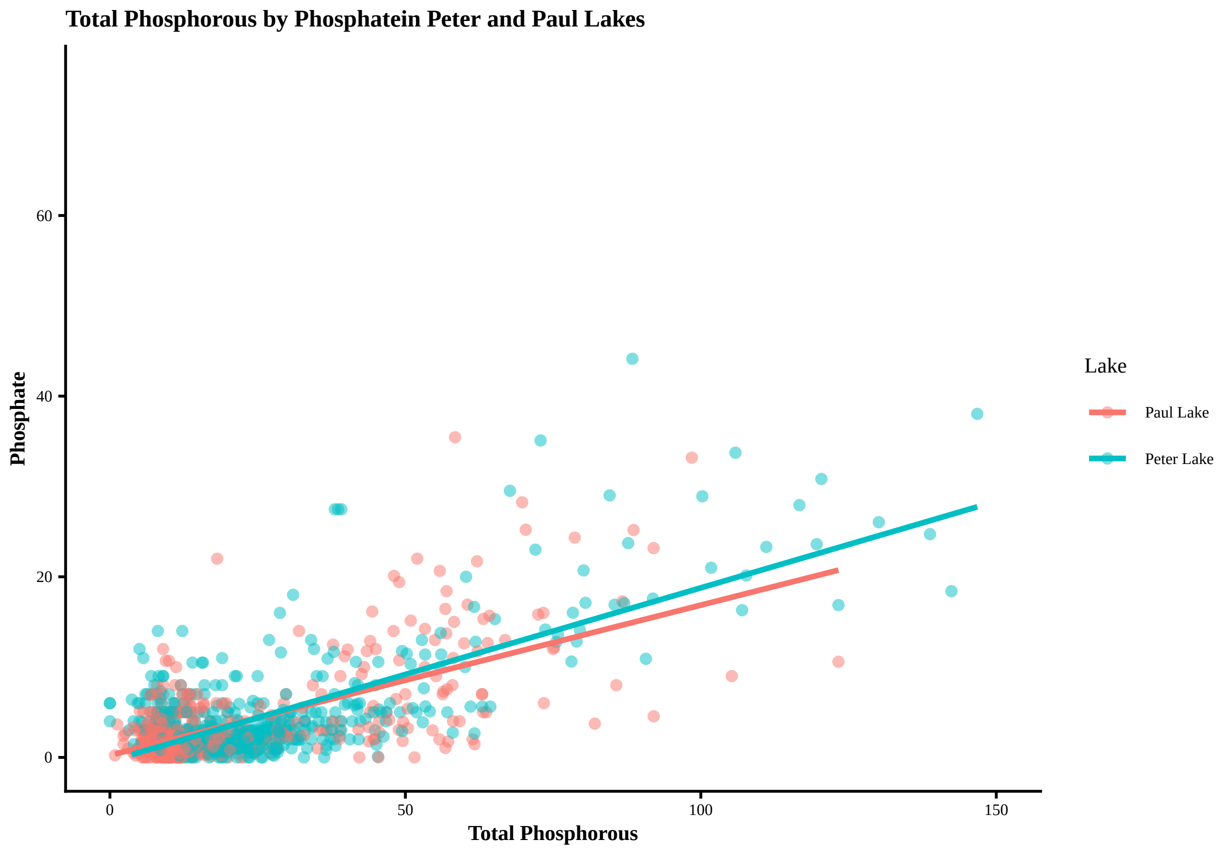
```
#4
NTL_LTER_processed_Chem_nutr %>%
  ggplot(aes(x = tp_ug, y = po4, color = lakename)) +
  geom_point(alpha = 0.5) +
  labs(x = "Total Phosphorous",
       y = "Phosphate") +
  xlim(0, 150) +
  ylim(0, 75) +
  labs(color = "Lake",
       title = "Total Phosphorous by Phosphatein Peter and Paul Lakes"
       ) +
  geom_smooth(method = lm,
              se = F) +
  my_theme01
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 21948 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21948 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_smooth()`).
```



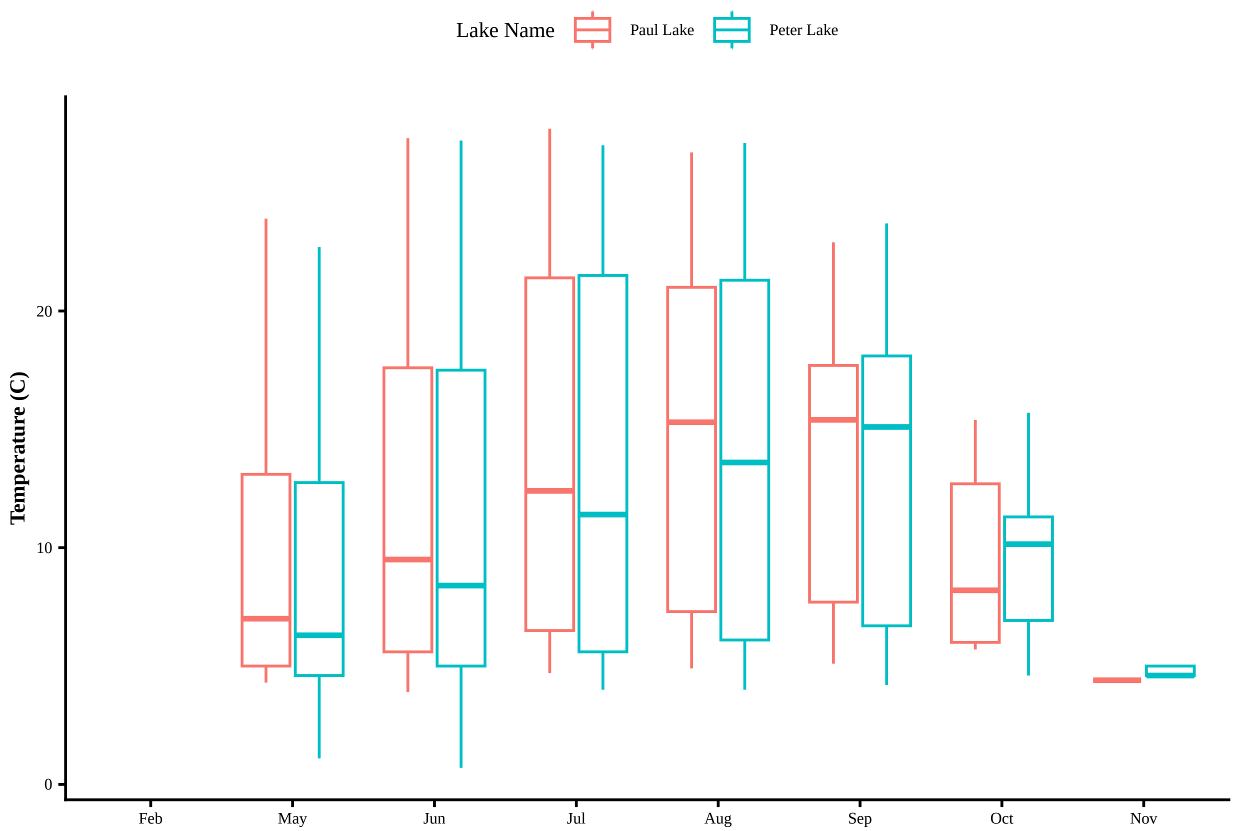
5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
NTL_LTER_processed_Chem_nutr$month <- month(NTL_LTER_processed_Chem_nutr$month, label = TRUE, abbr = TRU
#boxplot for temperature
NTL_LTER_processed_temp <-
  ggplot(NTL_LTER_processed_Chem_nutr) +
  geom_boxplot(aes(x = month,
                   y = temperature_C,
                   color = lakename)) +
  labs(x = element_blank(),
       y = "Temperature (C)",
       color = "Lake Name") +
  my_theme01 +
  theme(legend.position = "top")
print(NTL_LTER_processed_temp)
```

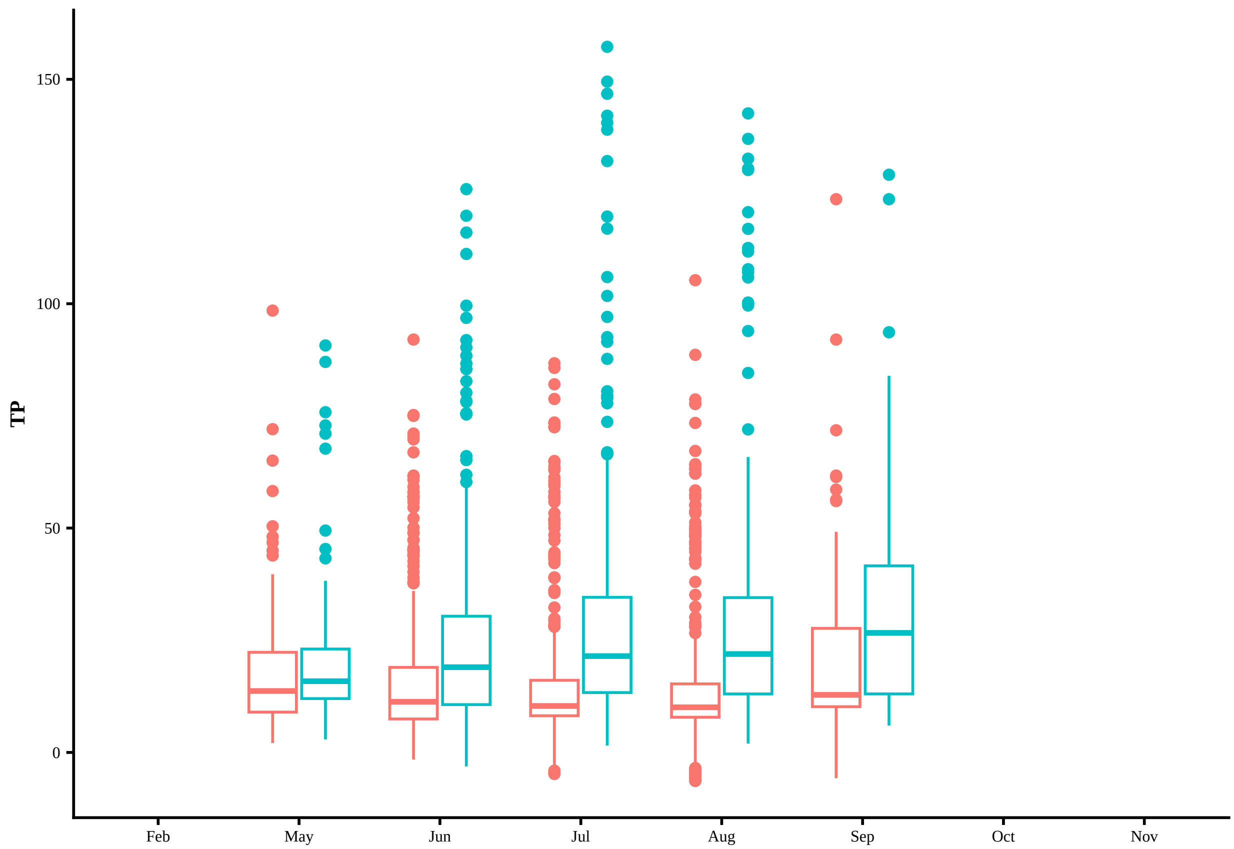
```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: 'label' cannot be a <ggplot2::element_blank> object.
```



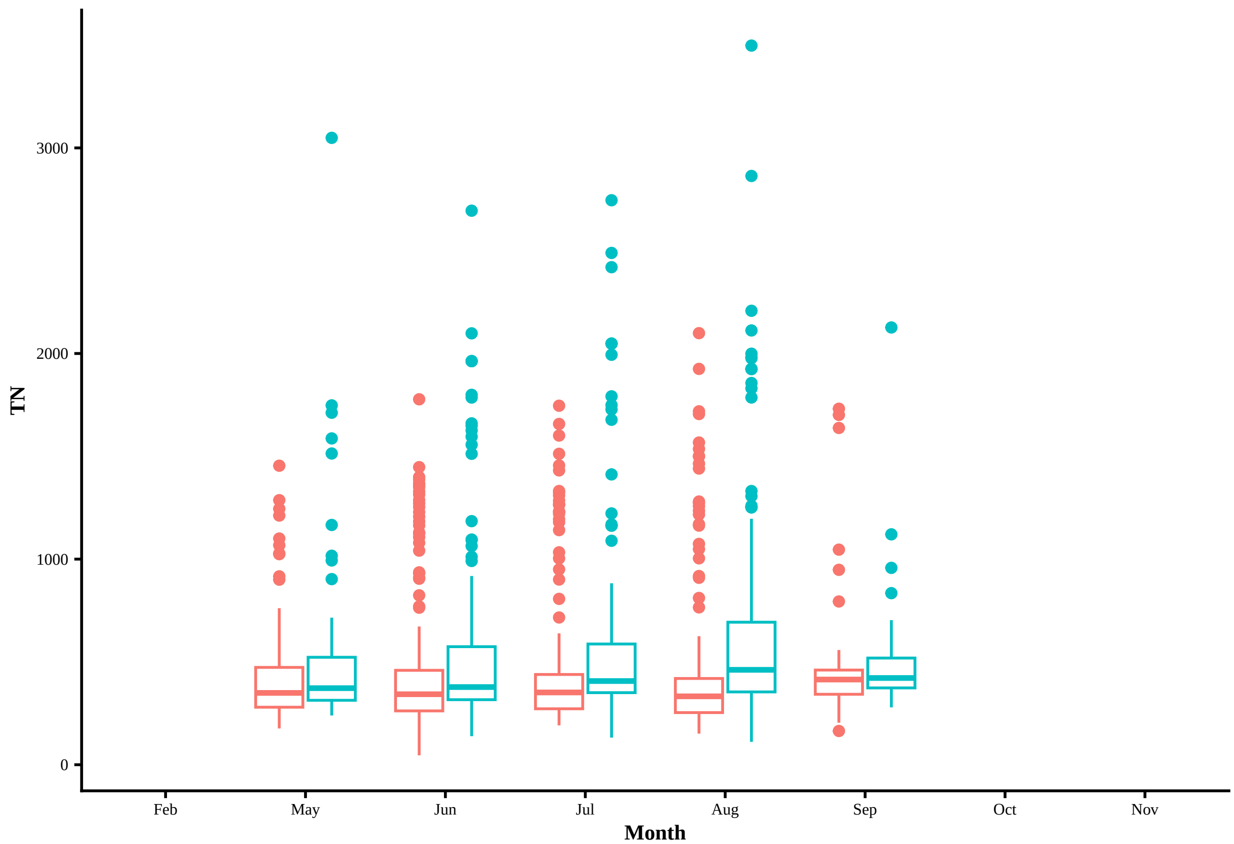

```
#boxplot for TP
NTL_LTER_processed_TP <- ggplot (NTL_LTER_processed_Chem_nutr) +
  geom_boxplot(aes(x = month,
                   y = tp_ug,
                   color = lakename)) +
  labs(x = element_blank(),
       y = "TP") +
  my_theme01 +
  theme(legend.position = "none")
print(NTL_LTER_processed_TP)
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').
## 'label' cannot be a <ggplot2::element_blank> object.
```




```
#boxplot for TN
NTL_LTER_processed_TN <- ggplot (NTL_LTER_processed_Chem_nutr) +
  geom_boxplot(aes(x = month,
                   y = tn_ug,
                   color = lakename)) +
  labs(x = "Month",
       y = "TN") +
  my_theme01 +
  theme(legend.position = "none")
print(NTL_LTER_processed_TN)
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```




```
#cowplot?
combined_plot <-
  plot_grid(NTL_LTER_processed_temp,
            NTL_LTER_processed_TP,
            NTL_LTER_processed_TN,
            ncol = 1,
            rel_heights = c(1, 1, 1))
```

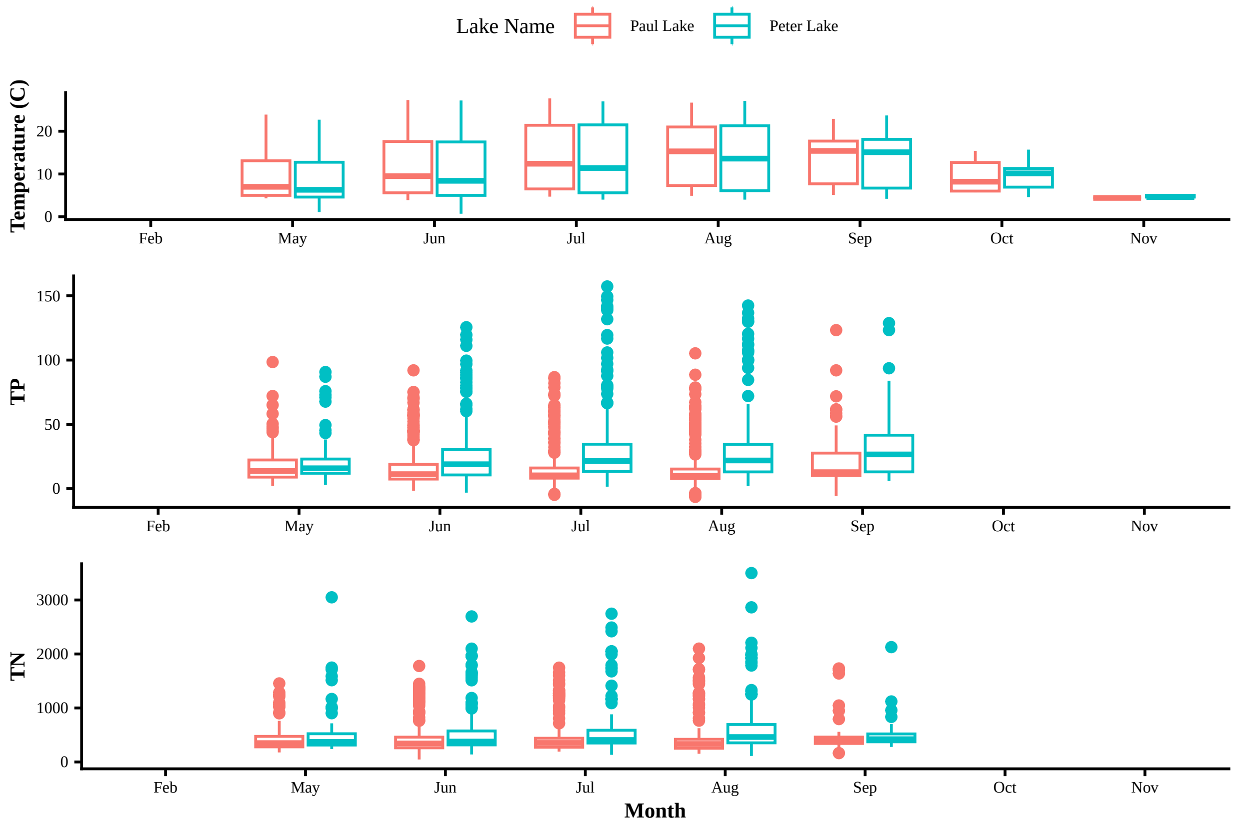
```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
## ‘label‘ cannot be a <ggplot2::element_blank> object.

## Warning: Removed 20729 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).

## Warning: ‘label‘ cannot be a <ggplot2::element_blank> object.

## Warning: Removed 21583 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

```
print(combined_plot)
```

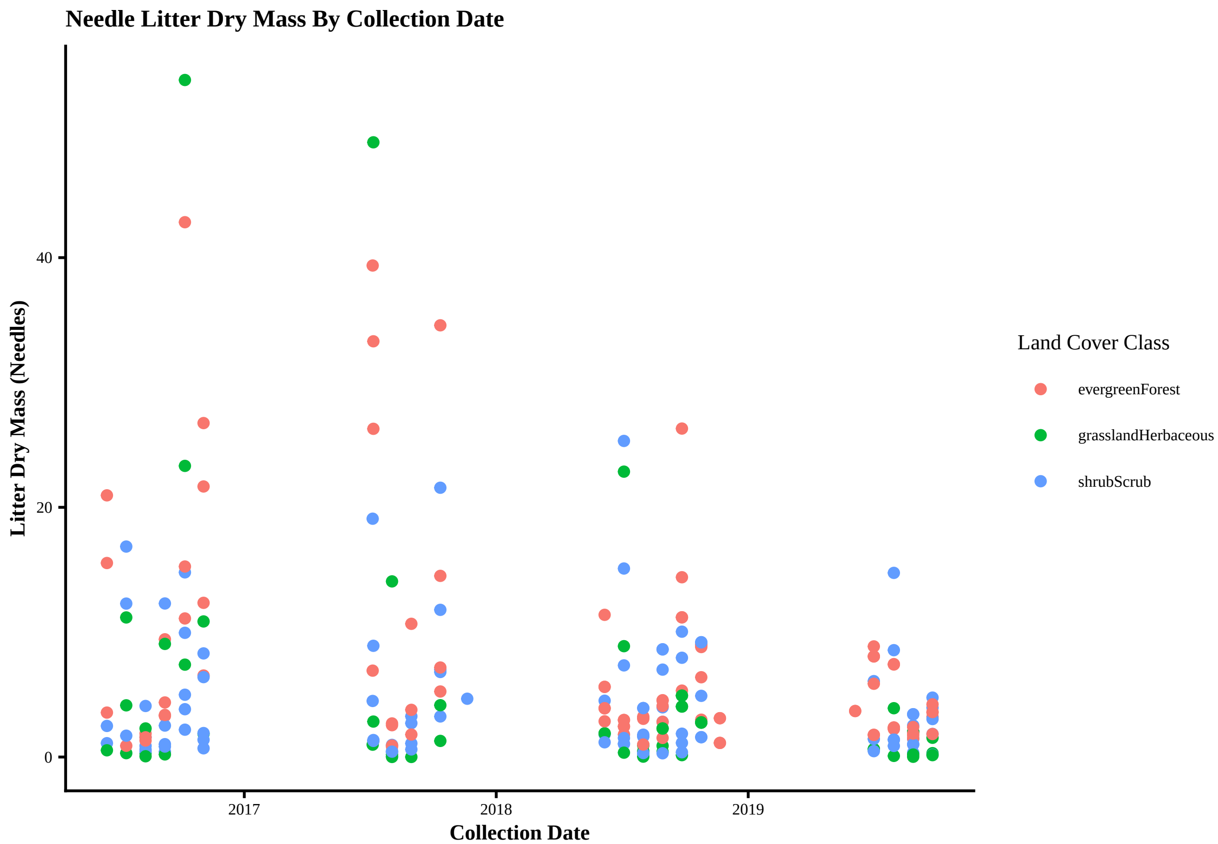


Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: For both lakes there is a temperature increase across the summer months going into fall, which is to be expected. Paul Lake has a higher temperature increase from May to August, but then decreases temperature in October, below Peter Lake's average temperature. Paul Lake has has lower amounts of TP and TN compared to Peter Lake during all months observed. Peter Lake increases in the aount of TP from May to September.

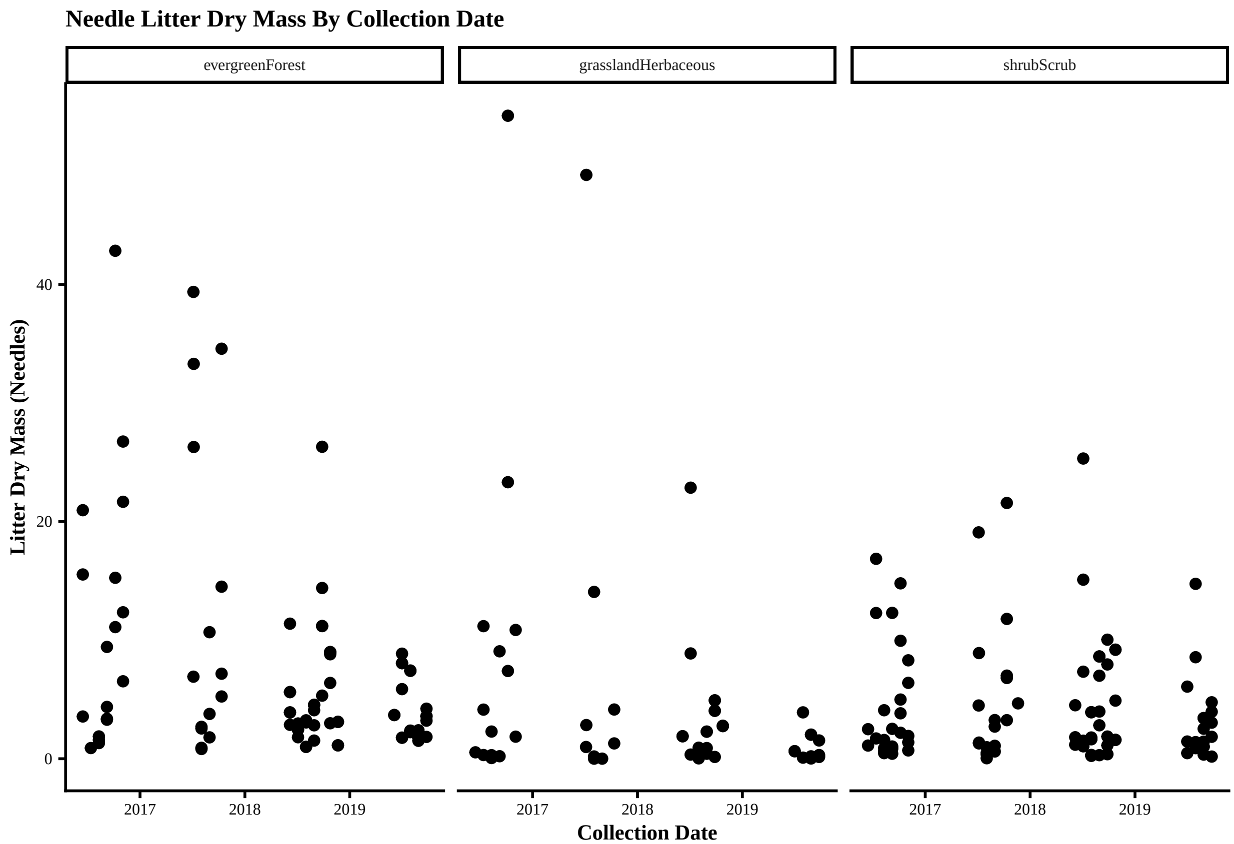
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
NEON_NIWO_processed_Litter_mass_trap %>%
  filter(functionalGroup == "Needles") %>%
  ggplot() +
  geom_point(aes(x = collectDate,
                 y = dryMass,
                 color = nlcdClass)) +
  labs(title = "Needle Litter Dry Mass By Collection Date",
       x = "Collection Date",
       y = "Litter Dry Mass (Needles)",
       color = "Land Cover Class") +
  my_theme01
```




```r
#7
NEON_NIWO_processed_Litter_mass_trap %>%
  filter(functionalGroup == "Needles") %>%
  ggplot() +
  geom_point(aes(x = collectDate,
                 y = dryMass)) +
  facet_wrap(~ nlcdClass) +
  labs(title = "Needle Litter Dry Mass By Collection Date",
       x = "Collection Date",
       y = "Litter Dry Mass (Needles)",
       color = "Land Cover Class") +
  my_theme01
```

```
## Ignoring unknown labels:
## * colour : "Land Cover Class"
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think that the second plot (number 7) is more effective. I think that it is more effective because it is easier to see where the data lies for each of the land cover types individually and thus able to compare them better. While it is nice when the data is colored by land cover, it is harder to read in the parts where the data overlaps each other. I think that a combination of creating facets for each land cover type and coloring by land cover type would be useful to clearly analyze the data. The different colors would help distinguish between the land cover types when looking across the plot. Or maybe even coloring by year so that it is easy to compare between the collection years. It just depends on what the research/analysis question is and what is being studied.